

# Zebra FX9600 Fixed UHF RFID Reader

The Zebra FX9600 is a high-performance fixed UHF RFID reader designed for high-volume, demanding environments. With up to 8 highly sensitive monostatic RF ports, IP53 die-cast aluminum housing, support for PoE/PoE+, and on-reader integrated applications, it delivers industry-leading RFID performance for warehouse, loading dock, and supply chain logistics.



## PHYSICAL CHARACTERISTICS

|                          |                                                           |
|--------------------------|-----------------------------------------------------------|
| Dimensions               | 27.3 cm L x 18.4 cm W x 5 cm D                            |
| Weight                   | Approx. 4.4 lb / 2.13 kg                                  |
| Housing Material         | Die-cast aluminum, IP53                                   |
| Visual Status Indicators | Multicolor LEDs: Power, Activity, Status and Applications |

## RFID SPECIFICATIONS

|                         |                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Max Receive Sensitivity | -86 dBm monostatic                                                                                                                                |
| Air Protocol            | ISO 18000-63 (EPC Class 1 Gen 2 V2)                                                                                                               |
| Frequency (UHF Band)    | Global: 902-928 MHz (and partial-band countries), 865-868 MHz; USA-only: 902-928 MHz                                                              |
| Transmit Power Output   | 0 dBm to +33.0 dBm (PoE+, 24V Ext. DC, Universal 24 VDC PSU); 0 dBm to +31.5 dBm (PoE, 12V Ext. DC for 4-port, 24V Ext. DC, Universal 24 VDC PSU) |

## CONNECTIVITY

|                     |                                                                              |
|---------------------|------------------------------------------------------------------------------|
| Connections         | 10/100 BaseT Ethernet (RJ45); USB Host & Client (Type A & B); Serial (DB9)   |
| General Purpose I/O | 4 in, 4 out, optically isolated (terminal block)                             |
| Power               | PoE (802.3af); PoE+ (802.3at); +24V DC (UL approved)                         |
| Antenna Ports       | FX9600-4: 4 monostatic ports (RP-TNC); FX9600-8: 8 monostatic ports (RP-TNC) |

ENVIRONMENTAL

|                       |                                      |
|-----------------------|--------------------------------------|
| Operating Temperature | -4 °F to +131 °F / -20 °C to +55 °C  |
| Storage Temperature   | -40 °F to +158 °F / -40 °C to +70 °C |
| Humidity              | 5% to 95% non-condensing             |
| Housing               | IP53                                 |

HARDWARE, OS AND FIRMWARE

|                      |                                                                                  |
|----------------------|----------------------------------------------------------------------------------|
| Processor            | Texas Instruments AM3505 (600 MHz)                                               |
| Memory               | Flash 512 MB; DRAM 256 MB                                                        |
| Operating System     | Linux                                                                            |
| Firmware Upgrade     | Web-based and remote firmware update capabilities                                |
| Management Protocols | RM 1.0.1 (HTTP / HTTPS via XML and SNMP); RDMP                                   |
| Network Services     | DHCP, HTTPS, FTPS, SFTP, SSH, HTTP, FTP, SNMP, NTP                               |
| Network Stack        | IPv4 and IPv6                                                                    |
| Security             | Transport Layer Security v1.2, FIPS-140                                          |
| API Support          | Host applications – .NET, C and Java EMDK; Integrated applications – C, Java SDK |

COMPLIANCE

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| Safety     | UL 60950-01, UL 2043, IEC 60950-1, EN 60950-1                                             |
| RF/EMI/EMC | FCC Part 15, RSS 210, EN 302 208, ICES-003 Class B, EN 301 489-1/3; Malaysia: 919-923 MHz |
| SAR/MPE    | FCC 47CFR2:OET Bulletin 65; EN 50364                                                      |
| Other      | RoHS, WEEE                                                                                |

FEATURE HIGHLIGHTS

- High RF sensitivity for superior read range and accuracy
- Durable die-cast aluminum housing with IP53 rating
- Supports PoE and PoE+ for simplified installation
- Up to 8 antenna ports per reader (FX9600-8 model)
- On-reader integrated applications for edge analytics
- Wi-Fi & Bluetooth connectivity via dongle
- Optional GPS-capable 4G LTE cellular modem

PRODUCT GROUP

|               |                       |
|---------------|-----------------------|
| Category      | RFID Reader           |
| Product Group | Fixed UHF RFID Reader |